magnified things by a factor of fifty, but the accuracy of his drawings also owes much to his painstaking approach. Hooke would make numerous sketches from many different angles before combining them into a single image.

Although it is not known for certain who developed the first microscopes, they were certainly in use by the 1660s. The early instruments were unreliable—due to the difficulty of making the lenses—and scientists had to be inventive and work around the problem. At first, Hooke had difficulty seeing his specimens clearly, so he invented an improved light source, named a “scotoscope.”

Hooke’s book is more than just an accurate representation of what he saw through the lens; it also theorizes on what the images reveal about the workings of the organisms he studied. For example, when looking at a wafer-thin specimen of cork, Hooke saw a honeycomb-like pattern, the elements of which he described as “cells”—a term that is still used today.

“… in every little particle… we now behold almost as great a variety of Creatures, as we were able before to reckon up in the whole Universe itself.”

Robert Hooke

Microscopic marvels

Micrographia inspired many other scientists to investigate the microscopic world. Following notes and diagrams from Hooke’s book, Dutch scientist Antonie van Leeuwenhoek was able to construct his own microscopes. He achieved magnifications of more than 200 times actual size.

Van Leeuwenhoek examined samples of rainwater and stagnant pondwater and marveled at the multitude of life he saw there. He identified single-celled protozoa, naming them “animalcules,” and went on to discover bacteria. He also made many observations of human and animal anatomy, including blood cells and sperm.